Sensor Report

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***Abstract*—An autonomous Formula Student Driverless (FSD) racecar requires a tightly synchronized hierarchy of exteroceptive sensors to perceive track boundaries and traffic cones alongside proprioceptive and by-wire feedback sensors to estimate vehicle dynamics, longitudinal and lateral tire slip, and actuator states. High-speed autonomous racing subjects instrumentation to extreme physical limits—including 1.5g to 2.5g lateral accelerations, high-frequency structural chassis vibrations, intense electromagnetic interference (EMI) from the 80 kW / 400–600 V high-voltage tractive system, and rapidly varying outdoor illumination. This technical paper presents an in-depth engineering compilation of the complete FSD sensor suite: the Hesai Pandar40P 3D LiDAR, Stereolabs ZED 2i stereo camera, VectorNav VN-200 GPS-aided Inertial Navigation System (INS), comparative solid-state transducer physics (capacitive, piezoresistive, and piezoelectric accelerometers; Hall-effect, AMR/TMR, and fluxgate magnetometers), optical sensors (encoders and ground speed), and active Hall-effect wheel speed sensors. Furthermore, hardware integration on the NVIDIA Jetson AGX Orin central compute platform, electrical wiring, CAN-FD architecture, mathematical error mitigation pipelines, and current Indian market sourcing strategies are rigorously established.**

***Keywords*—Formula Student Driverless (FSD), Transducer Physics, Hall-Effect, Extended Kalman Filter, Sensor Fusion, NVIDIA Jetson AGX Orin, Autonomous Vehicles.**

I. Introduction

The autonomous hardware pipeline of a Formula Student Driverless (FSD) racecar follows a strict sequential and closed-loop hierarchy: Sensing → Perception → Path Planning → Motion Control running atop the Robot Operating System (ROS 2) middleware [1]. Because no standalone transducer can maintain high reliability across all track conditions, the vehicle instrumentation is partitioned into functional categories: exteroceptive environment perception, proprioceptive state estimation, and by-wire actuation feedback [2]. This report details the theoretical working principles of these sensors, the physics underlying their transduction mechanisms, strategies for vehicle integration and error mitigation, and sourcing options within the Indian market.

II. Vehicle Integration: Wiring, Mounting, and Power

Integrating highly sensitive low-voltage (LV) sensors inside a chassis housing an 80 kW high-voltage (HV) electric powertrain requires stringent wiring, routing, and power isolation protocols.

- **Mechanical Mounting:** The LiDAR is mounted atop a rigid mast on the main roll hoop to provide a 360° unobstructed view. The INS is rigidly bolted directly to the chassis near the Center of Gravity (CG) to minimize lever-arm centripetal acceleration offsets. Cameras are secured inside weatherproof enclosures pitched downward to avoid sky glare.
- **Power Isolation and Distribution:** All logic controllers and sensors are powered by an independent 24 V LV battery isolated completely from the 400–600 V HV tractive system. Galvanically isolated DC-DC step-down converters supply clean 12 V and 5 V rails to the sensors.
- **EMI Shielding and Wiring Routing:** To prevent induced noise from the HV system, LV sensor harnesses are strictly segregated from HV power cables. Where crossings are unavoidable, LV and HV cables cross at 90° right angles. Analog signals and Ethernet cables utilize shielded twisted pair (STP) wiring with the shield grounded at only one end to prevent ground loops.
- **Isolated CAN-FD Bus Architecture:** The compute unit and Vehicle Control Unit (VCU) exchange deterministic steering and braking commands over a robust 1 Mbps CAN-FD bus using galvanically isolated CAN transceivers featuring 120 Ω termination resistors at extreme ends of the network.

III. Exteroceptive Sensors

Exteroceptive sensors perceive the external track environment to localize traffic cones with centimeter-level spatial accuracy [1].

A. 3D Mechanical LiDAR: Hesai Pandar40P

The Hesai Pandar40P is a 40-channel mechanical rotating laser scanner operating on the Direct Time-of-Flight (dToF) principle. Pulsed laser diodes emit infrared pulses, and Avalanche Photodiodes (APDs) detect the returning photons. Radial range d is calculated via:

$$d = (c\,\Delta t) / 2 \tag{1}$$

where c is the speed of light and Δt is the round-trip propagation delay. Unlike uniform LiDARs, the Pandar40P concentrates 25 channels at an ultra-fine vertical resolution of 0.33° near the horizon. This guarantees that multiple scan rings hit small 325 mm track cones out to 30–40 m, enabling robust Euclidean clustering.

B. Stereo Vision: Stereolabs ZED 2i

The Stereolabs ZED 2i solves monocular depth ambiguity by employing dual CMOS imagers separated by a rigid horizontal baseline B = 120 mm. Given the horizontal disparity d = u_L - u_R between matched features, metric depth Z is recovered via triangulation:

$$Z = (f\,B) / d \tag{2}$$

Depth uncertainty σ_Z grows quadratically with distance (σ_Z ∝ Z²), which demonstrates why stereo cameras excel in near-field dense color classification but require LiDAR for long-range precision. The ZED 2i can be equipped with an optional add-on linear polarizer to suppress asphalt specular glare.

C. Optical Sensors: Encoders, Photodiodes, and Ground Speed

Beyond standard cameras, optical sensing elements drive multiple critical FSD sub-systems. Optical rotary encoders on the steering column use an LED shining through a precision slotted disc onto a quadrature photodiode array, producing A/B phase-shifted square waves to determine precise angular position [2]. Because driven and braked tires experience longitudinal slip, optical flow ground speed sensors are used to measure true vehicle velocity independent of tire dynamics. While professional motorsport uses reference-grade sensors like the Kistler Correvit SFII (which costs lakhs of rupees), Formula Student teams employ cost-effective downward-facing optical flow modules that track asphalt surface micro-texture displacements to compute true v_x and v_y.

TABLE I
Weighted Decision Matrix (Pugh Trade-Off) for Core Perception and Navigation Sensors

Subsystem	Selected Model	Evaluated Alternatives	Key Selection Justification and Quantitative Engineering Trade-Off
3D LiDAR	Hesai Pandar40P (40-Channel)	Velodyne VLP-16, Ouster OS1-32, Livox Mid-360	0.33° dense horizon beam spacing (25 channels in −6° to +2°) hits 325 mm cones at 35 m; 200 m range; ±2 cm accuracy beats VLP-16 (2.0° gap) and Mid-360 (non-repetitive scan).
Stereo Vision	Stereolabs ZED 2i (2.1 mm)	Intel RealSense D455, Dual Basler Mono Rig	120 mm rigid baseline (vs. 95 mm on D455) reduces depth uncertainty (σ_Z ∝ Z²); passive neural stereo works in bright sunlight; optional polarizer for glare.
INS / IMU	VectorNav VN-200 (Tactical MEMS)	Xsens MTi-670G, Bosch BNO085, InvenSense MPU9250	<5°/hr gyro in-run bias stability and <0.04 mg accel stability; 400 Hz EKF output; 1 PPS sync. Consumer IMUs (MPU9250) drift >10 m in 5 s due to high bias instability.

IV. Proprioceptive Sensors and Transducer Physics

Proprioceptive sensors measure the internal kinematic state of the vehicle body [2].

A. IMU vs. AHRS vs. INS: The VectorNav VN-200

A standalone Inertial Measurement Unit (IMU) outputs raw 3-axis accelerometer and gyroscope data at high rates (e.g., 800 Hz). An Attitude and Heading Reference System (AHRS) adds a magnetometer to output 3D orientation. In contrast, the VectorNav VN-200 is a full GPS-Aided Inertial Navigation System (INS). It integrates an IMU, magnetometer, and GNSS receiver (running at 5 Hz) with an onboard Extended Kalman Filter (EKF) running at 400 Hz.

In pure dead-reckoning, uncompensated accelerometer bias b_a causes position error to grow quadratically ($\delta p_a \propto \frac{1}{2} b_a t^2$). However, uncompensated gyroscope bias b_g causes an orientation error that erroneously projects gravity g into the horizontal plane. Double-integrating this false gravity projection causes position error to grow cubically:

$$\delta p_g(t) \approx (1/6)\, g\, b_g\, t^3 \quad (3)$$

The VN-200 bounds this cubic drift by fusing the high-rate 800 Hz inertial data with 5 Hz absolute GNSS updates inside its 400 Hz EKF, continuously estimating and subtracting the biases.

B. Comparative Physics of Accelerometers

- **Capacitive MEMS (Used in VN-200):** Dynamic acceleration causes a proof mass to shift, modulating differential capacitance between interdigitated fingers. It responds down to 0 Hz (true DC), allowing it to measure static gravity for pitch/roll inclination and sustained cornering g-forces [2].
- **Piezoelectric Accelerometers:** A seismic mass compresses a PZT ceramic element, generating an electrostatic charge. Because the charge leaks away under static loads, it cannot measure 0 Hz DC gravity; it is strictly restricted to high-frequency chassis vibration diagnostics [2].
- **Piezoresistive Accelerometers:** Utilizes doped silicon strain gauges in a Wheatstone bridge. It measures DC to high frequencies and excels in high-shock (>500g) environments but exhibits higher thermal drift [2].

C. Comparative Physics of Magnetometers

- **Hall-Effect Magnetometers:** When a bias current I flows through a semiconductor slab exposed to a perpendicular magnetic field B, the Lorentz force deflects charge carriers, generating a transverse Hall voltage [2]:

$$V_H = (I B) / (n q t) \quad (4)$$

where n is charge carrier density, q is elementary charge, and t is thickness. Due to lower sensitivity, they are poor compasses but ideal for wheel speed sensors [2].

- **Magnetoresistive (AMR / TMR):** Thin films change electrical resistance based on the angle between the internal magnetization and external magnetic field. Highly sensitive (nano-Tesla resolution), they are the standard compasses inside tactical INS units like the VN-200.
- **Fluxgate Magnetometers:** Consists of high-permeability ferromagnetic cores driven into saturation by AC coils. Highly accurate but too bulky and power-hungry for FS packaging [2].

TABLE II
Comparative Solid-State Physics of Accelerometer and Magnetometer Transducers

Transducer Type	Governing Physical Law / Equation	Freq. / Range	Primary Advantages	Limitations and FSD Suitability
Capacitive MEMS Accel.	$\Delta C \approx (2\varepsilon A/d_0^2)\, x$; electrostatic force balance	0 Hz (DC) to 1 kHz; ±16g	True DC gravity response; low thermal drift; high bias stability [1].	Susceptible to VRE if un-isolated; Standard in VN-200 INS.

Piezoresistive Accel.	$\Delta R/R_0 = G_F \varepsilon$ (silicon strain-gauge bridge)	0 Hz (DC) to 5 kHz; up to 2000g	High shock survivability; simple DC Wheatstone readout [1].	High thermal drift (TCR); used for high-G impact logging [1].
Piezoelectric Accel.	$Q = d_{33} F = d_{33} m a$ (PZT/quartz charge)	1 Hz to 20 kHz (No DC)	Wide dynamic range; self-generating; ultra-high bandwidth [1].	Charge leaks at 0 Hz; cannot measure static tilt or steady cornering g [1].
Hall-Effect Mag.	$V_H = (I B) / (n q t)$ (Hall effect)	DC to 25 kHz; mT to T range	Monolithic CMOS integration; immune to saturation at high B [1].	Low sensitivity for Earth 50 µT field; used in wheel speed and pedals [1].
AMR / TMR Mag.	$R(\theta) = R_\perp + \Delta R \cos^2\theta$ (AMR model)	DC to 1 MHz; nT to mT range	Micro-Tesla resolution; compact 3-axis bridge; low noise.	Vulnerable to EV tractive current EMI; requires Hard/Soft-Iron cal.
Fluxgate Mag.	$V_{2f} \propto B_{ext}$ (saturable-core second harmonic)	DC to 100 Hz; pT to µT	Ultra-low offset drift; aerospace and marine reference grade [1].	Bulky wound coils; high power; unsuitable for compact FSD packaging [1].

D. Wheel Speed: Inductive Proximity vs. Active Hall-Effect

Passive inductive sensors generate voltage proportional to the rate of change of magnetic flux ($V \propto d\Phi/dt$). At low speeds, the signal amplitude drops to zero, making them unsuitable for launch control. Industrial inductive proximity sensors (e.g., Autonics PR12-4DN) detect metal teeth via LC oscillator dampening. However, FSD racecars typically prefer Active Hall-Effect Gear-Tooth ICs (e.g., Allegro ATS series or Honeywell SS series). These contain integrated bias magnets, differential Hall elements, and Schmitt triggers, outputting clean digital square waves all the way down to 0 Hz (true standstill) [2].

V. By-Wire Actuation and FS Rules Compliance

Closed-loop Steer-by-Wire (SbW) and Brake-by-Wire (BbW) control loops require high-bandwidth feedback sensors that strictly comply with Formula Student rules for System Critical Signals (SCS). Steering rack travel is measured via linear potentiometers, while hydraulic brake pressure is monitored using piezoresistive diaphragm sensors (e.g., Bosch 0 261 545 053).

Per SCS rules, these sensors are configured with an offset live-zero analog range of 0.5 V to 4.5 V. Any reading <0.25 V (open circuit/short to ground) or >4.75 V (short to supply) immediately triggers an SCS fault, opening the shutdown circuit and engaging the fail-safe Emergency Brake System (EBS).

VI. Error Mitigation Strategies

High-voltage tractive systems in FSD cars (e.g., 80 kW peak power at 400–600 V) draw peak currents of 130 A to 200 A. These high-voltage cables generate intense magnetic flux densities ($B = \mu_0 I / 2\pi r$) that completely overwhelm Earth's ~50 µT magnetic field. Because this dynamic tractive current B_{EMI} cannot be statically calibrated out, magnetometer heading updates are often rejected or heavily down-weighted in the EKF during dynamic driving [1].

Furthermore, longitudinal wheel slip during acceleration corrupts odometry. This is mitigated by fusing wheel speed data with INS linear acceleration using a Mahalanobis distance gating test in the EKF, rejecting wheel encoders whenever tire slip spikes.

TABLE III
Failure Modes, Physical Noise Mechanisms, and Hardware/Software Mitigation Matrix

Sensor Subsystem	Primary On-Track Failure / Error Mode	Hardware Mitigation Strategy	Software / Algorithmic Mitigation Strategy
Hesai Pandar40P LiDAR	100 ms rotational scan skew at 15 m/s; pitch ground hits; rain/dust noise.	Rigid roll-hoop mast mount; dual-return optical receiver hardware.	400 Hz IMU point-by-point SE(3) deskewing; RANSAC/Patchwork++ ground removal; 1D histogram filter [1].
Stereolabs ZED 2i Camera	Quadratic depth error ($\sigma_Z \propto Z^2$); asphalt specular glare; vibration blur.	120 mm baseline + 4 mm lens; optional linear polarizer; Sorbothane isolator.	Lower-ROI auto-exposure metering; LiDAR-Camera frustum fusion beyond 12 m [1].
VectorNav VN-200 INS	Cubic dead-reckoning drift ($\propto t^3$); Vibration Rectification Error (VRE); HV EMI.	CoG vibration-damped aluminum mount; >150 mm separation from HV busbars.	Onboard EKF GPS/IMU tight coupling; disable magnetometer yaw under HV load; Allan variance tuning.
Wheel Speed Sensors	0 V signal dropout at standstill (inductive); wheel lockup or spin slip error.	Active back-biased Hall-Effect sensor + Schmitt trigger; 1.0 mm air-gap jig.	Mahalanobis innovation gating in EKF; cross-check against Optical Ground Speed sensor.
SbW & BbW Actuation	Wire break, short circuit, or EMI voltage spike causing unintended actuation.	Shielded twisted-pair wiring; 0.5–4.5 V offset range or 4–20 mA current loop.	FS Rule SCS window comparator (<0.25 V or >4.75 V triggers instant EBS shutdown).

VII. Central Compute, Vehicle Integration, and Time Sync

A. NVIDIA Jetson AGX Orin (64 GB) Compute Architecture

It is essential to distinguish that the NVIDIA Jetson AGX Orin is not a sensor, but the Central AI Compute and Sensor Fusion Server [1]. Delivering up to 275 TOPS of heterogeneous compute across an NVIDIA Ampere GPU, DLAs, and Cortex-A78AE CPU, the Orin executes the entire autonomous software stack within a strict <35 ms glass-to-actuator latency budget. The GPU runs quantized YOLO cone detection on camera frames and Euclidean clustering on point clouds, while the CPU executes the 400 Hz EKF and MPC control laws [1].

B. Hardware Time Synchronization: 1 PPS and IEEE 1588 PTP

At a racing velocity of 20 m/s (72 km/h), a mere $\Delta t = 25$ ms clock offset between the camera and LiDAR produces a 0.50 m spatial registration error. To achieve sub-millisecond temporal alignment, the VN-200 GNSS receiver disciplines its internal oscillator to UTC satellite time and outputs a hardware 1 PPS (Pulse Per Second) signal. This 1 PPS pulse is routed into both the Hesai Pandar40P Interface Box and a hardware GPIO interrupt pin on the Jetson AGX Orin. The Orin synchronizes the LiDAR's internal clock via IEEE 1588 Precision Time Protocol (PTP) and triggers global-shutter camera exposures via GPIO pulses, locking all ROS 2 header stamps to a single microsecond timebase.

TABLE IV
Master FSD Sensor and Compute Suite Architecture, Sampling Rates, and Bus Interfaces

Subsystem Layer	Selected Hardware Model	Primary Role in FSD Pipeline	Sampling Rate	Electrical / Data Interface
3D Range Perception	Hesai Pandar40P (40-Ch LiDAR)	3D point cloud, cone centroid detection (0.3–50 m), LiDAR odometry.	10 Hz / 20 Hz	100BASE-TX GbE UDP + 1 PPS / IEEE 1588 PTP
Stereo & Mono Vision	Stereolabs ZED 2i + Global Shutter Mono	RGB cone color classification, dense near-field depth (0.5–15 m).	30–60 Hz	USB 3.0 Type-C (Screw-Lock) + GPIO Trigger
Inertial Nav & GNSS	VectorNav VN-200 (GPS-Aided INS)	6-DoF specific force & angular rate, EKF attitude, global GNSS pose.	800 Hz (IMU) / 5 Hz (GNSS)	RS-232 / TTL UART (921.6 kbps) + 1 PPS Out
Wheel & Ground Speed	Active Hall-Effect WSS (×4) + Optical Flow	Wheel angular velocity ω_i, longitudinal slip ratio λ, true v_x, v_y.	100–500 Hz	MCU Timer Schmitt Trigger → Isolated CAN-FD
By-Wire Feedback	Optical Encoder, Linear Pot, Bosch Pressure (×2)	Closed-loop SbW rack angle δ and BbW/EBS hydraulic pressure (0–250 bar).	200–1000 Hz	0.5–4.5 V Ratiometric / SSI → VCU ADC → CAN-FD
Central AI Compute	NVIDIA Jetson AGX Orin (64 GB, 275 TOPS)	YOLOv8 TensorRT inference, LiDAR-Camera fusion, EKF, SLAM, MPC [1].	Real-time (<35 ms loop)	GbE, USB 3.2, Dual MTTCAN CAN-FD, UART, GPIO

VIII. Indian Market Sourcing Strategy and Cost

Procuring specialized racing hardware in India requires navigating customs duties and local availability. High-value sensors like the Hesai Pandar40P and VectorNav VN-200 are typically imported using university Department of Scientific and Industrial Research (DSIR) certificates (Notification 51/96-Customs), which exempts Basic Customs Duty (BCD) and restricts IGST to 5% for educational research. Microcontrollers, Hall ICs, and cameras are procured in INR via local distributors. Table V details the estimated Bill of Materials (BOM) and verified Indian suppliers as of October 2026.

TABLE V
Categorized Bill of Materials (BOM), Realistic INR Cost Estimates, and Suppliers (Verified Oct 2026)

Sensor / Component	Selected Model	Est. Cost (INR)	Verified Indian Suppliers / Procurement Channel
3D Mechanical LiDAR	Hesai Pandar40P	₹4.5L – ₹7.5L*	Corebolt LLP (Hyderabad); Academic Direct Import via DSIR.
Stereo Depth Camera	Stereolabs ZED 2i	₹96,599 – ₹1,03,549	Robu.in (Pune); ThinkRobotics India. (Polarizer optional)
GPS-Aided INS / IMU	VectorNav VN-200	₹2.8L – ₹4.2L*	Kusaba Engineers / Tirumala Aerospace (Pune) or Direct DSIR Import.
Central AI Compute	NVIDIA Jetson AGX Orin 64GB	₹1,85,000 – ₹2,15,000	Robu.in (Pune); Evelta Electronics (Mumbai); Arrow India.
Inductive Proximity Sensor	Autonics PR12-4DN	₹550 – ₹800 / unit	Robu.in; Evelta; Local automation stockists.
Active Hall-Effect Sensor	Allegro ATS / Honeywell SS	₹300 – ₹800 / IC	Mouser India; DigiKey India; RS Components India.
Optical Ground Speed	Downward Optical Flow Chip	₹18,000 – ₹45,000	Evelta / Robu.in. (Correvit SFII alternative).
BbW Brake Pressure	Bosch 0 261 545 053 (250 bar)	₹1,200 – ₹16,500	Authorized Bosch Dealers / Boodmo; Mouser India.

* High-value import estimates reflect academic pricing and University DSIR customs duty exemption (Notification 51/96-Customs). Prices sourced October 2026 via listed distributor catalogs.

IX. Conclusion

A reliable autonomous Formula Student architecture balances solid-state transducer physics against harsh vehicular constraints. Fusing the LiDAR's non-uniform point cloud with stereo vision eliminates perception blind spots, while the INS explicitly bounds cubic dead-reckoning drift. Adhering to hardware synchronization and strict SCS rules ensures robust, fail-safe racing performance.

X. References

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